Governed Enterprise Intelligence Assistant

Topic 12 – Governed, Optimized Workflow

1. Project Overview

The **Governed Enterprise Intelligence Assistant** is an AI-powered enterprise reporting workflow developed using Google Opal.

The project demonstrates how enterprise data can be processed through independent AI generation tasks, compiled into an executive review portal, monitored through structured execution logs, and reviewed through a human-in-the-loop approval process.

The workflow follows the **5:3:1 governance model**, emphasizing monitoring, controlled authorization, and human oversight.

2. Objectives

- Accept enterprise data as input.
- Execute independent AI generation tasks.
- Generate structured enterprise reports.
- Extract important business metrics and trends.
- Identify business insights, risks, and opportunities.
- Generate monitoring and audit information.
- Compile outputs into an executive review portal.
- Validate explicit manager approval decisions.
- Prevent unauthorized saving authorization.

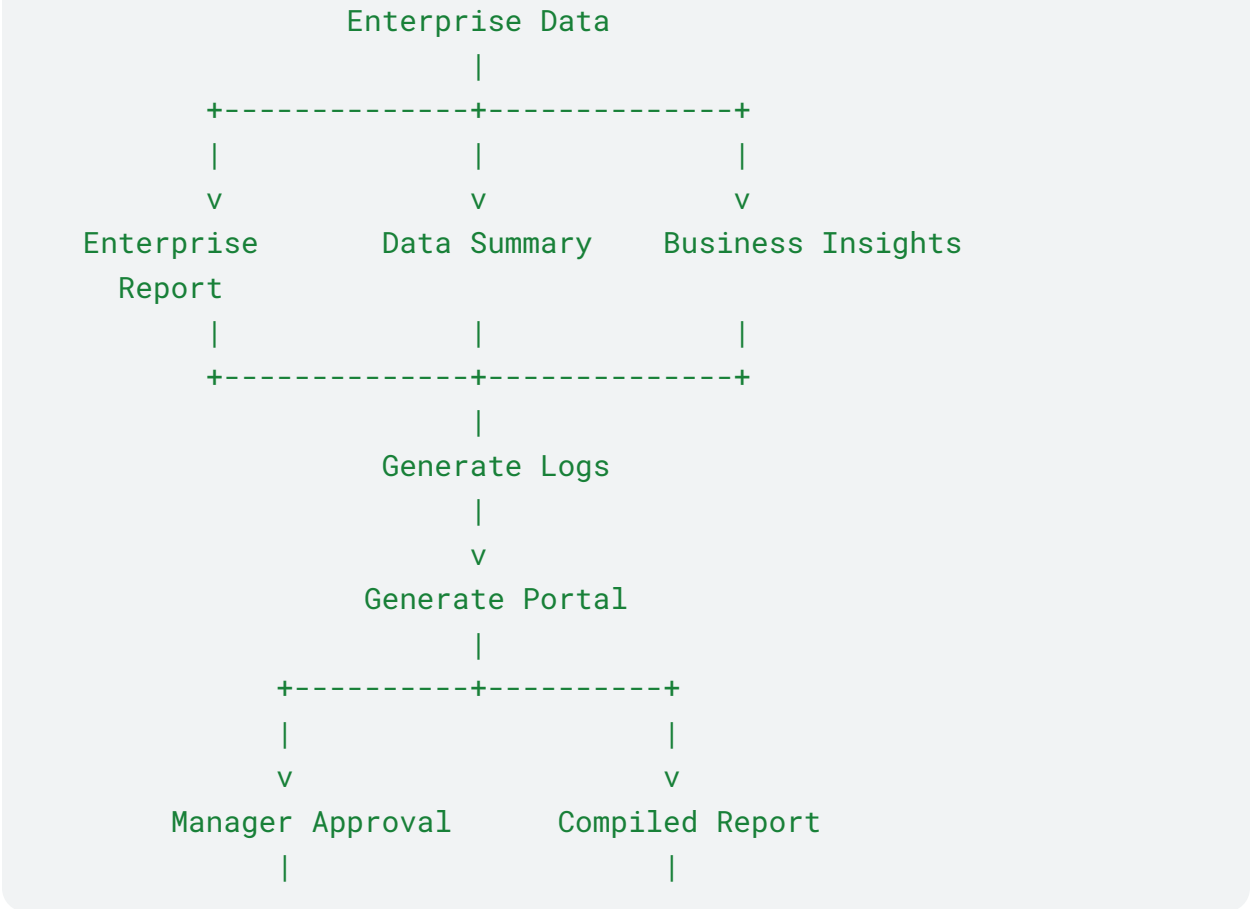
3. Technology Used

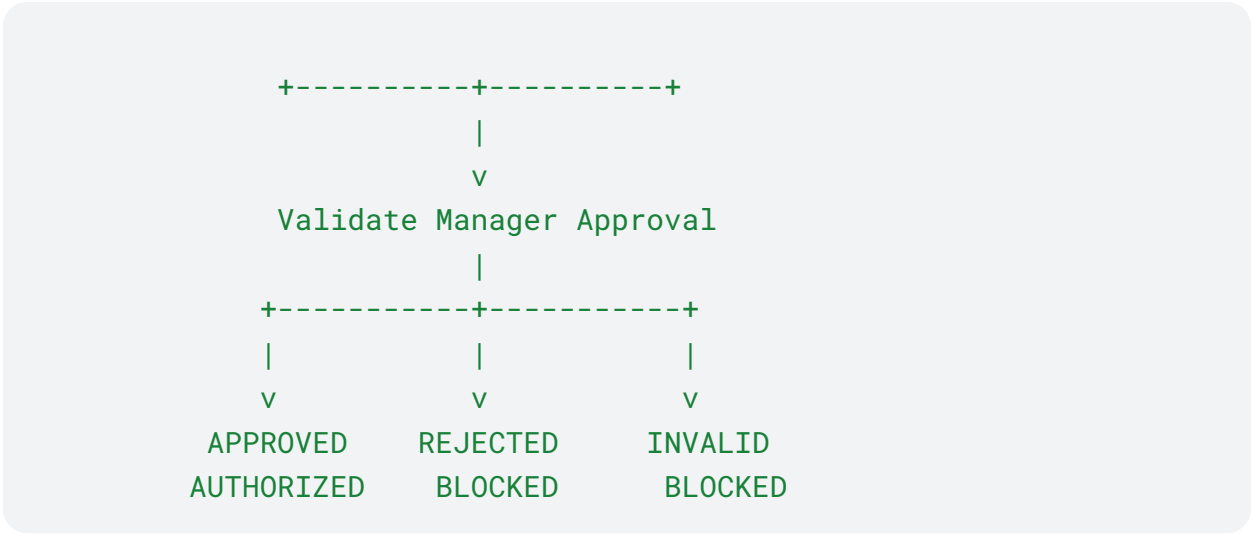
Technology	Purpose
Google Opal	Workflow development and execution
Generative AI	Enterprise report generation
AI Data Summarization	Extracting important metrics
AI Business Analysis	Identifying insights and opportunities
Opal Memory	Supporting available workflow memory functionality
Markdown	Project documentation

4. Workflow Architecture

The workflow accepts enterprise data and distributes it to independent generation and monitoring tasks.

None





Note: This diagram illustrates the logical workflow. Automatic conditional routing and Google Drive saving are not implemented in the current version.

5. Main Workflow Components

Component	Responsibility
Enterprise Data	Accepts the source enterprise document.
Generate Enterprise Report	Creates a structured business report.
Generate Data Summary	Extracts important figures and trends.
Extract Business Insights	Identifies opportunities, risks, and recommendations.
Generate Logs	Prepares workflow monitoring information.
Generate Portal	Compiles outputs into an executive review portal.
Manager Approval	Collects the manager's explicit decision.
Validate Manager Approval	Validates the decision and determines authorization status.

6. 5:3:1 Governance Model

5 – Monitoring

Monitoring provides visibility into workflow execution.

The Generate Logs component prepares information such as:

- Workflow name.
- Execution start timestamp.
- Enterprise input received.
- Individual generation task statuses.
- Execution completion timestamp.
- Available execution latency.
- Errors and warnings.
- Overall workflow status.

3 – Actuation

Actuation refers to controlled operations that modify or save enterprise information.

Examples include:

- Saving approved reports.
- Updating authorized business records.
- Publishing approved business outputs.

Only designated controlled steps should perform these operations.

Current implementation: Approval validation determines whether a saving operation may be authorized. Automatic saving is not yet configured.

1 – Human-in-the-Loop

Human oversight ensures that a manager explicitly reviews the generated report.

The manager must provide one of the following decisions:

- APPROVE
- REJECT

The validation agent checks the decision before any controlled saving operation can be authorized.

7. Manager Approval Validation

The approval validation component follows these rules:

Manager Input	Approval Status	Saving Authorization

APPROVE	APPROVED	AUTHORIZED
REJECT	REJECTED	BLOCKED
MAYBE	INVALID	BLOCKED
Blank	INVALID	BLOCKED

The system must never:

- Assume manager approval.
- Convert a rejected decision into an approved decision.
- Authorize invalid decisions.
- Claim that a report has been saved without confirmed storage success.

8. Sample Enterprise Input

The project uses fictional demonstration data for:

Company: GreenTech Solutions Pvt. Ltd.
Reporting Period: October 2026

Metric	Sample Value
Revenue	₹500,000
Expenses	₹350,000
Net Profit	₹150,000
Net Profit Margin	30%
Sales Growth	12%
Customer Retention	85%
Customer Satisfaction	4.3/5

The source data is used to verify the accuracy of generated reports and summaries.

9. Executive Review Portal

The generated executive portal contains the following sections:

1. Executive Summary.
2. Financial Performance.
3. Customer Performance.
4. Operational Performance.
5. Business Insights.
6. Monitoring and Audit Log.
7. Human Approval Status.

The portal provides a consolidated view of generated enterprise information for manager review.

10. Testing Results

Four approval-validation test cases were executed.

Test ID	Test Scenario	Input	Expected Result	Status
TC-01	Valid approval	APPROVE	APPROVED / AUTHORIZED	PASS
TC-02	Manager rejection	REJECT	REJECTED / BLOCKED	PASS
TC-03	Invalid decision	MAYBE	INVALID / BLOCKED	PASS
TC-04	Blank approval	Empty	INVALID / BLOCKED	PASS

Additional workflow verification covered:

- Independent generation connections.
- Executive portal input connections.
- Manager approval connections.
- Executive portal sections.
- Enterprise data accuracy.
- Monitoring output.
- Pending approval status.
- Prevention of false storage confirmation.

Approval Validation: 4/4 tests passed.

11. Security and Governance

The project applies the following governance practices:

- Restricted workflow sharing using available Opal settings.
- Explicit manager approval.

- Validation of approval decisions.
- Blocking of rejected and invalid decisions.
- Transparent reporting of unavailable integrations.
- No false claims of completed saving operations.

Full enterprise RBAC enforcement has not been independently verified.

12. Known Limitations

The following features remain pending:

1. **Google Drive Integration:** Automatic saving of approved reports to shared company Google Drive is not implemented because the required saving tool was unavailable in the current Opal editor.
2. **Conditional Routing:** Automatic branching based on approval status is not implemented because the required conditional tool was unavailable.
3. **Centralized Audit Storage:** Persistent centralized audit-log storage has not been confirmed.
4. **Enterprise RBAC:** Full role-based access control enforcement has not been independently verified.
5. **Console Debugging:** Complete Opal Console debugging has not been verified.

These limitations are documented to ensure transparency about the current implementation.

13. Future Enhancements

- Integrate an approved Google Drive saving action.
- Implement approval-based conditional routing.
- Configure persistent centralized audit storage.
- Strengthen enterprise role-based access control.
- Add automated workflow failure notifications.
- Perform complete end-to-end integration testing.

14. Project Documentation

The repository should contain the following documents:

```
None
Governed-Enterprise-Intelligence-Assistant/
│
├── README.md
```

```
│
├── concept_guide.md
│
├── requirement_document.md
│
├── implementation_plan.md
│
├── testing_results.md
│
└── prompts/
    ├── generate_logs.md
    ├── generate_enterprise_report.md
    ├── generate_data_summary.md
    ├── extract_business_insights.md
    ├── generate_portal.md
    └── validate_manager_approval.md
```

15. Conclusion

The Governed Enterprise Intelligence Assistant demonstrates the core principles of optimized enterprise workflows through independent AI generation tasks, structured reporting, monitoring, and human-in-the-loop approval validation.

The four approval-validation scenarios passed successfully.

The project provides a foundation for responsible enterprise AI workflow development using the 5:3:1 governance model.

Final Project Status: CORE TESTS PASSED – PARTIAL IMPLEMENTATION

Project: Governed Enterprise Intelligence Assistant
Topic: 12 – Governed, Optimized Workflow
Platform: Google Opal
Prepared for: Trainer Demonstration